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# An Open-Source NuMAD Model for the IEA 15 MW Blade with Baseline Structural Analysis

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Open-source reference models are highly informative and valuable for advancing wind energy technology as usage of these models by the research community enables opportunities for innovation of new designs and technology solutions. This work details an open-source reference model created with the Numerical Manufacturing and Design (NuMAD) tool of the International Energy Agency (IEA) 15 MW open-source reference wind turbine blade, and documents the detailed blade structure and baseline structural performance. NuMAD is a widely used tool in blade design, and this work provides one more resource for the wind energy community. Detailed high-fidelity structural analyses (modal, deflection, strength, fatigue, buckling, and aero-elastic stability) are performed using the University of Texas at Dallas (UTD) IEA 15 MW blade model following international design standards of blade design to baseline the blade structural performance. We provide a detailed description of important assumptions of Wind-Plant Integrated System Design and Engineering Model (WISDEM) and NuMAD, which are two different blade modeling tools, that were used to create the blade in NuMAD format based on the original IEA 15 MW blade design specifications. The detailed documentation of the open-source UTD IEA 15 MW blade increases the body of knowledge on large-scale turbines, serves as reference for future studies and research, and exemplifies the detailed structural design following industry standards for blade design.

## I. Nomenclature

$MW$  = Megawatt  
 $rpm$  = Revolutions per minute  
 $V_r$  = Rated wind speed  
 $\mu\epsilon$  = Microstrain

## II. Introduction

THE open-access designs of a complete wind turbine system are often known as Reference Wind Turbines (RWTs), which provide supporting models for simulation and design, a platform for performance and cost evaluations, and propose the design modifications before prototype development. These open-source models provide the industry, researchers, and students a tool for designing the wind turbines of the future. International Energy Agency (IEA) 15 MW [1], Technical University of Denmark (DTU) 10 MW [2], and National Renewable Energy Laboratory (NREL) 5 MW [3] are some of the examples of the RWTs.

The main objective of this work is to generate an open-source blade modeled with the Numerical Manufacturing and Design (NuMAD) tool based on the IEA 15 MW reference turbine [1] that increases the body of knowledge on large-scale turbines and that can serve as reference for future studies and research. Previously, work was performed in Ref. [1] to develop the IEA 15 MW offshore reference wind turbine using the Wind-Plant Integrated System Design and Engineering Model (WISDEM) [4], which is a turbine size that follows the industry trend of turbine size growth as shown by the recent installation of the turbine prototypes General Electric GE 14 MW Haliade-X and the Siemens-Gamesa 14

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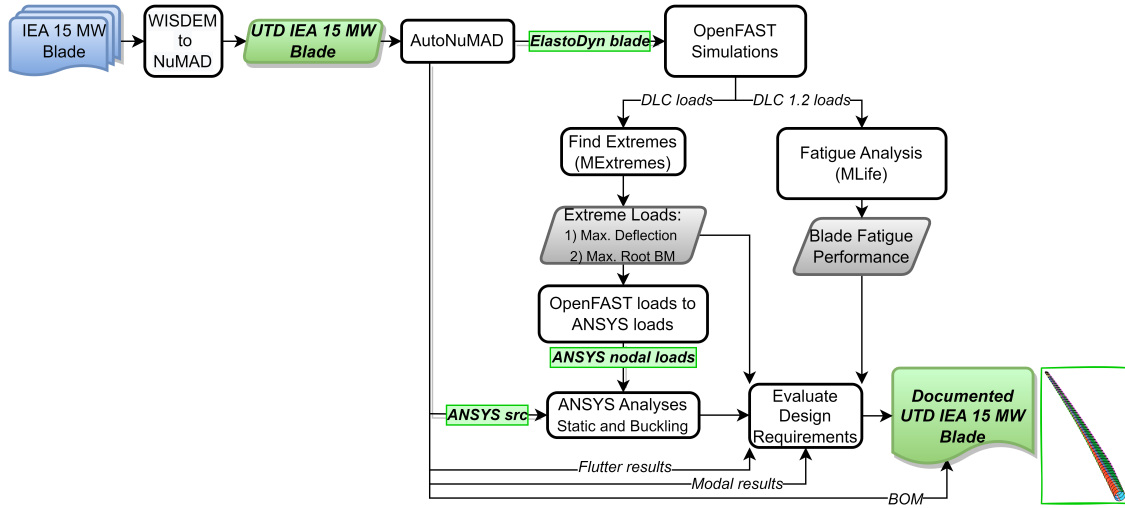
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MW SG 14-222 DD<sup>§</sup>. NuMAD is a tool developed by Sandia National Laboratories (SNL) with which a detailed blade geometry, material properties and lay-ups can be defined. Additionally, NuMAD has the capability to create a detailed shell blade model that can be evaluated with high-fidelity ANSYS Mechanical APDL software [5]. Having a NuMAD model of the IEA 15 MW blade is of great value to the research community because NuMAD is a widely used tool in blade design and due to the analysis capabilities it provides for detailed blade structural design studies for the composite layup and blade geometry. Accordingly, this paper presents research work involving the conversion process of the IEA 15 MW blade WISDEM format to NuMAD format.

The main structural analyses performed to meet the international design standards for blade design are deflection, modal, strength (strain/stress), fatigue, buckling, and aero-elastic stability [6, 7]. However, the main limitations found in the public report of the IEA 15 MW turbine are that blade fatigue, buckling, and aero-elastic stability were not evaluated, and no high-fidelity tools were used to determine the strain distributions on the blade. As mentioned by the authors of the IEA 15 MW turbine [1], while this blade design meets strength and frequency requirements, it may not meet all the blade structural requirements and a detail assessment is needed. Thus, in this work we present a full baseline structural evaluation of the (University of Texas at Dallas) UTD IEA 15 MW blade model including modal, deflection, strength, fatigue, buckling, and aero-elastic stability analyses following international design standards of blade design [6, 7]. All the assumptions and results are fully documented and the model is publicly available in GitHub to the wind energy research community for their use and reference [8].

### III. Methodology

The approach used to create the UTD IEA 15 MW blade and perform detailed structural analyses is illustrated in Fig. 1. Detailed structural analyses include modal, deflection, fatigue, buckling, and aero-elastic stability (flutter).



**Fig. 1 UTD IEA 15 MW blade model creation and analysis process.**

The software used in the design and analysis process are tabulated in Table 1. The UTD IEA 15 MW blade is modeled using AutoNuMAD [9–14] which is a semi-automated version of NuMAD v2.0 [9]. AutoNuMAD is connected to Precomp [15] and BModes [16] that are used to generate the OpenFAST<sup>¶</sup> blade file, OpenFAST (formerly know as FAST [17]) is used with TurbSim and IECWind to perform aero-elastic simulations with turbulent and non-turbulent wind files, MExtremes [18] is used to post-process the results and find a critical design load case (DLC), ANSYS Mechanical APDL [5] is used for high-fidelity strain and buckling performance evaluation, and MLife [19, 20] is used to assess the fatigue performance of the blade. The flutter performance is determined with the tool described in Ref. [21].

<sup>§</sup>Information obtained from websites: <https://www.ge.com/renewableenergy/wind-energy/offshore-wind/haliade-x-offshore-turbine> and <https://www.siemensgamesa.com/products-and-services/offshore/wind-turbine-sg-14-222-dd> (last accessed on May 30, 2022).

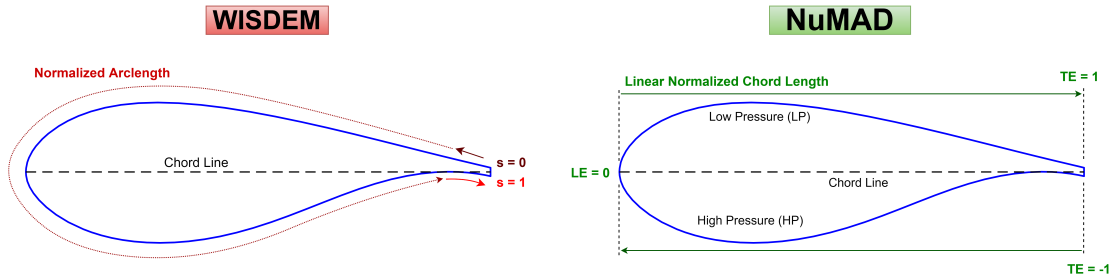
<sup>¶</sup>OpenFAST documentation: <https://openfast.readthedocs.io/en/main/> (last accessed on May 30, 2022).

**Table 1 Design and analysis software**

| Task                                                                 | Tools Used            |
|----------------------------------------------------------------------|-----------------------|
| Blade Design, Bill of Materials, and Aero-elastic Stability Analysis | AutoNuMAD             |
| Pre-processors                                                       | PreComp and Bmodes    |
| Aero-servo-elastic Analyses                                          | OpenFAST              |
| Post-processor                                                       | MExtremes and MLife   |
| High-Fidelity Analysis of Strength and Stability                     | ANSYS Mechanical APDL |

### A. Conversion of Blade Design from WISDEM to NuMAD

As illustrated in Fig. 1, the IEA 15 MW blade files <sup>‡</sup> (which include the public report, a yaml file with the wind turbine information for WISDEM, and an Excel spreadsheet summarizing the design) [1] were transformed to the Excel file format required by NuMAD. A thorough inspection of the documents was performed to determine a complete set of material properties, thickness and layer orientation of components with the same material (e.g. spar cap), number and location of unique regions on a cross-section defined by different lay-ups (e.g. leading edge reinforcement), and region widths. To transform the blade information to a NuMAD format, understanding the definition of division points and region widths along a cross-section for both WISDEM and NuMAD was necessary. Additionally, understanding of the lay-up sequence of materials during manufacturing is important. The following Fig. 2 shows how division points in a cross-section are defined in WISDEM and NuMAD.



**Fig. 2 Different methods (WISDEM and NuMAD) to define division points on a cross-section that determine the width of a region on the airfoil. Division points are defined based on arc-length for WISDEM and based on percent of chord for NuMAD (where negative values indicate the high-pressure side and positive values for the low-pressure side) [4, 9].**

In WISDEM, the different regions on an airfoil cross-section are defined by the arc-length start and end points [4]. The airfoils, regions on the airfoil cross-section, and the thickness of same-material stacks are defined for ten blade span stations <sup>\*\*</sup>. This detailed information is used to create the NuMAD blade following the steps listed below:

- Format airfoil coordinates to be readable by NuMAD. Then, the WISDEM normalized arc-length coordinates were transformed to normalized chord coordinates used by NuMAD using a MATLAB [22] script that first calculates the cumulative arc-length of the airfoil cross-section including the flatback, then interpolates the x and y coordinates of the WISDEM division points (defined by arc-length percent) using the cumulative arc-length and coordinates of the known airfoil.
- Only the x normalized coordinates are required as input to NuMAD. A negative value is assigned to the x coordinates to define NuMAD division points on the high-pressure side and positive for low-pressure side. The NuMAD division points are then organized in increasing order (from negative to positive with the leading edge (LE) being defined by a division point with value of zero).
- Add geometric details (chord and twist) of the last 1.3% of the blade span to create a smoother transition to the blade tip (which were not defined in the ten blade span stations) using information from the WISDEM yaml file

<sup>‡</sup> Information obtained from the release version number v1.0 and master branch: <https://github.com/IEAWindTask37/IEA-15-240-RWT>.

<sup>\*\*</sup> This information was found in the "Blade Shell Layup" tab of IEA-15-240-RWT\_tabular.xlsx spreadsheet in the master branch of the GitHub repository (release number v1.0).

and the Excel spreadsheet of the GitHub repository. Correctly modeling the blade tip is useful to obtain better estimates of blade mass and frequencies.

- Assign materials to the corresponding stacks based on the Excel spreadsheet and WISDEM yaml file available in the GitHub repository. Some ultimate tensile and compression properties of the materials in the master branch were not defined and seemed inconsistent (tri-axial glass properties having lower performance than bi-axial glass). A consistent and complete set of properties was found instead in the development branch (same elastic properties, but with appropriate ultimate strength properties). Fiber volume fraction (fvf) and fiber weight fraction (fwf) properties are also provided (which were used to create a bill of materials).

Mass, bill of materials, and frequency checks are performed to evaluate the NuMAD blade against IEA 15 MW blade results, and fine adjustments on the region widths are performed as necessary to minimize error.

## B. Detailed Structural Analyses

Detailed structural analyses are conducted to fully assess and document the UTD IEA 15 MW blade design performance (modes, deflection, strength, fatigue, buckling, and aero-elastic stability). The UTD IEA 15 MW blade model allows performing detailed structural analyses (fatigue, buckling, and aero-elastic stability) that were not previously performed in Ref. [1], and to check the strength with higher fidelity tools such as ANSYS Mechanical APDL. First, OpenFAST aero-servo-elastic simulations of relevant DLCs are performed assuming the turbine is onshore but at the coast (no waves). The DLC wind files are generated using IECWind and TurbSim following the International Electrotechnical (IEC) 61400-1 third edition standard [6]. The blade fatigue performance is determined using results of DLC 1.2, and buckling and strength are evaluated using DLC 1.4 and 1.5 results (which were shown to be the most critical cases for blade flapwise deflection and blade root flapwise bending moment in Ref. [1], respectively). The specifications of each DLC are shown in Table 2. The results of DLC 1.4 and DLC 1.5 are checked against the values reported in Ref. [1] to assure the blades are evaluated with similar wind conditions.

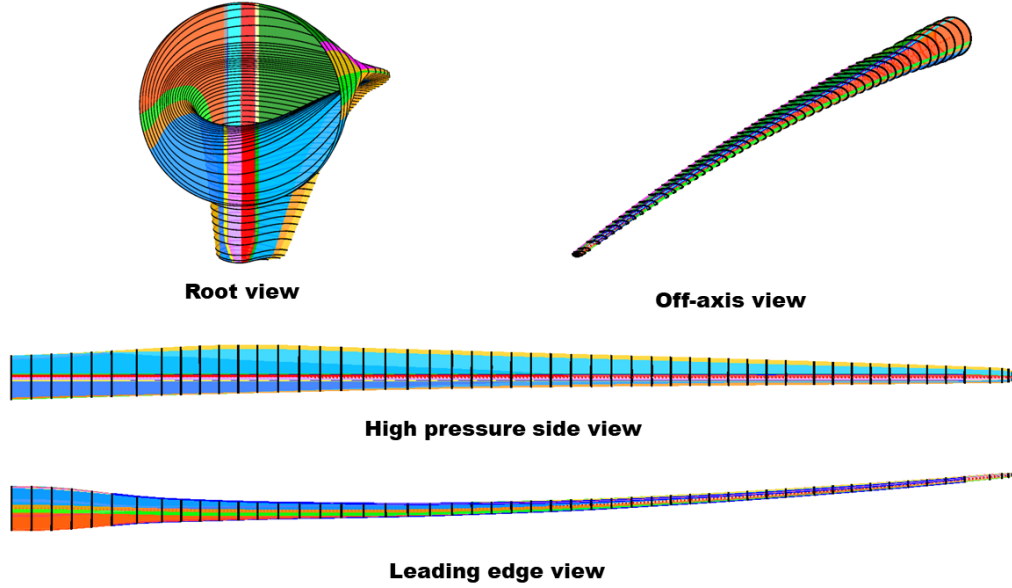
**Table 2 Design load cases analyzed [6, 23]**

|         |                                                                               |                                                                                                                                    |
|---------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| DLC 1.2 | Power production in normal turbulence<br>Normal turbulence model (NTM)        | Fatigue<br>3:2:25 m/s with six seeds per wind speed<br>100 s transient and 600 s runtime<br>Total simulations = 72                 |
| DLC 1.4 | Power production in extreme coherent<br>gust with wind direction change (ECD) | Ultimate strength<br>$V_r$ , $V_r \pm 2$ m/s with gust<br>140 s transient and 10 s runtime<br>Total simulations = 6                |
| DLC 1.5 | Power production in extreme wind shear (EWS)                                  | Ultimate strength<br>3:2:25 m/s $\pm$ Vert./Horiz. wind shear change<br>150 s transient and 12 s runtime<br>Total simulations = 48 |

The safety factors used to assess the structural performance of the blade design are obtained based on the IEC 61400-1 third edition [6] and the Germanischer Lloyd (GL) [7] standards. All partial safety factors for loads and composite materials are included to determine the following safety factors: 2.977 for strength, 2.042 for buckling and 1.642 for fatigue. Additionally, the blade tip deflection should result in at least 30% tower clearance in operational cases, a minimum fatigue life of 20 years must be met, and resonance should not occur. The safety factors, clearance and frequency requirements are consistent with studies by Ref. [21]. In this work, the strength performance is evaluated using the Von Mises failure criteria.

#### IV. UTD IEA 15 MW Blade Model

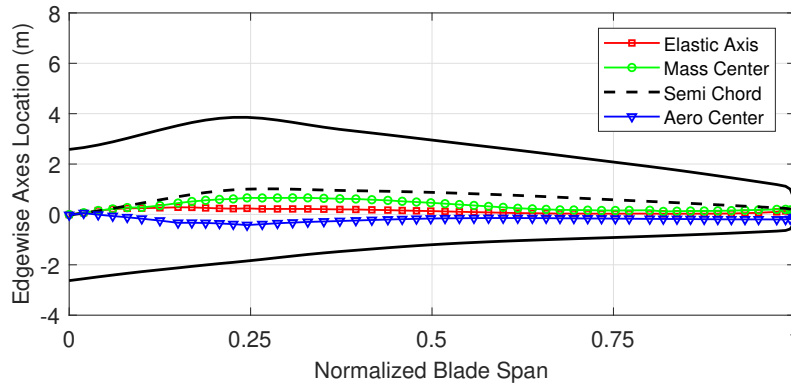
The 15 MW blade generated using NuMAD following the process described in Section III.A is shown in Fig. 3. The mass and modal properties of the UTD IEA 15 MW blade are presented in the following sub-sections. This blade is the direct conversion of the IEA 15 MW blade to NuMAD format, and does not include structural re-design.



**Fig. 3** UTD IEA 15 MW blade model with 53 blade span stations and 18 distinct lay-up regions along each blade cross-section.

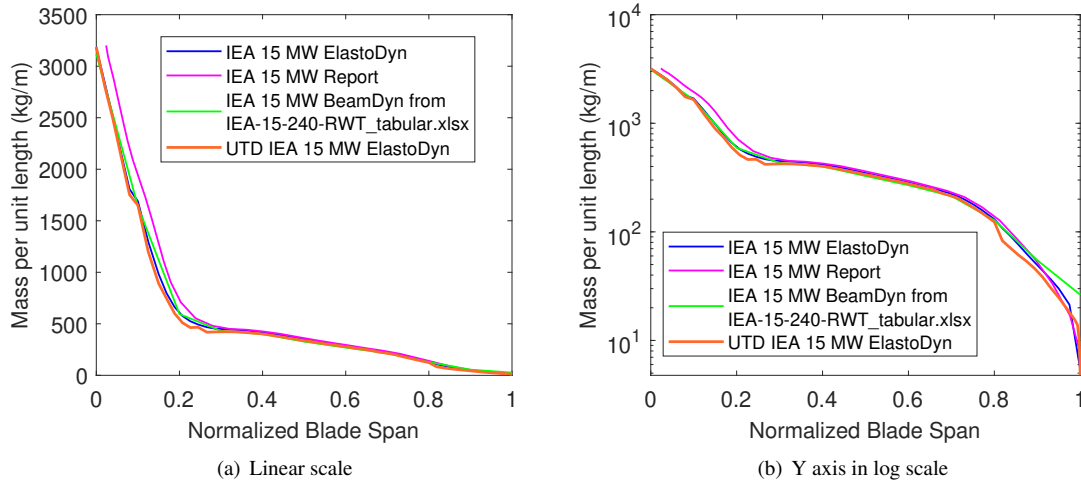
##### A. Distributed Properties

The blade planform and the location of the elastic, mass, and aerodynamic centers are shown in Fig. 4.



**Fig. 4** Blade planform and distribution of elastic, mass, and aerodynamic centers along the blade span.

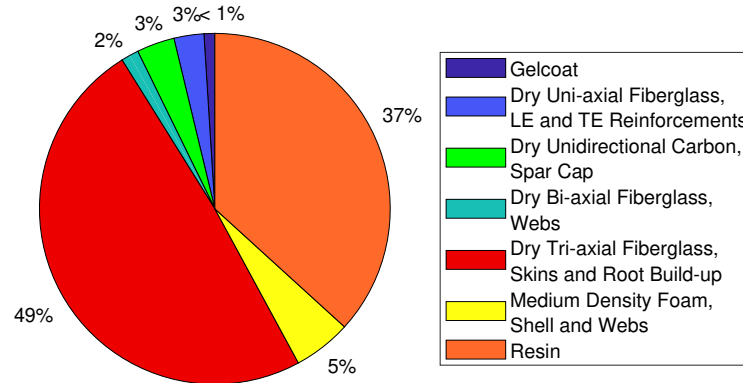
The mass per unit length (mass density) of the UTD IEA 15 MW blade modeled with NuMAD is shown in Fig. 5. The results were obtained with PreComp and are used in the ElastoDyn blade file. The UTD IEA 15 MW blade has an overall good match with the information of the IEA 15 MW blade. Information for the IEA 15 MW blade is available in multiple files in the same release, and there are some discrepancies between them for the mass per unit length. But the ElastoDyn properties for both the original IEA 15 MW blade and the UTD NuMAD version match well.



**Fig. 5 Mass per unit length of the UTD IEA 15 MW blade and comparison to IEA 15 MW information.**

### B. Mass and Bill of Materials (BOM)

AutoNuMAD is used to calculate the mass and bill of materials of the UTD IEA 15 MW blade. The mass of the UTD IEA 15 MW blade is 68,077 kilograms and this is 4.33% greater compared to blade model reported in Ref. [1]. Small differences in mass were expected given that some slight differences in the mass per unit length shown in Fig. 5. However, the higher mass of the UTD IEA 15 MW blade may be due to different mass estimation approaches used by AutoNuMAD and WISDEM. Figure 6 shows the mass composition of the UTD IEA 15 MW blade where tri-axial fiberglass and resin are the largest components.



**Fig. 6 Materials in UTD IEA 15 MW blade.**

### C. Modes

The first six blade modes at zero rpm (parked) obtained with BModes are shown in Table 3. In Ref. [1], it is stated that the blade design was mainly done with WISDEM which also uses BModes. Compared to blade model reported in Ref. [1] the first flapwise and edgewise frequencies of the UTD IEA 15 MW blade are 2.71% and 0.63% greater, respectively. These differences may be due to the different airfoil interpolation methods used by the different tools that

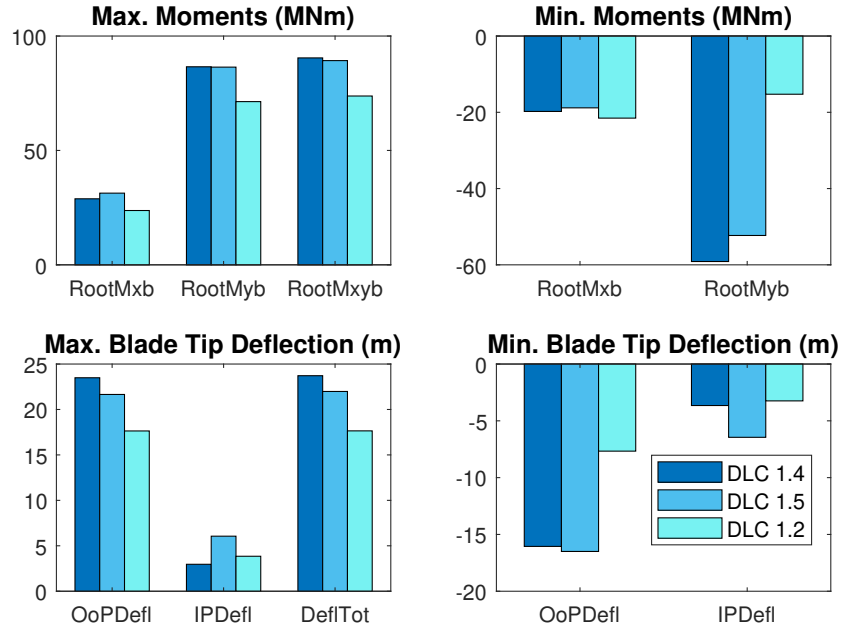
affects the sectional mass and stiffness properties. However, the UTD IEA 15 MW blade flapwise frequency is closer to the WISDEM output from running the IEA 15 MW blade yaml file (1.52% for flapwise and -2.09% for edgewise).

**Table 3 Blade parked modes**

| Mode | Frequency (Hz) | Description  |
|------|----------------|--------------|
| 1    | 0.57           | 1st Flapwise |
| 2    | 0.65           | 1st Edgewise |
| 3    | 1.72           | 2nd Flapwise |
| 4    | 2.08           | 2nd Edgewise |
| 5    | 3.41           | 3rd Flapwise |
| 6    | 4.29           | 1st Torsion  |

## V. Design Load Case (DLC) Analysis

The extreme blade root bending moments and tip deflections obtained from DLC 1.2, 1.4, and 1.5 OpenFAST simulations are shown in Fig. 7. In Ref. [1], the maximum root bending moments occur in DLC 1.4 and 1.5, and the maximum flapwise tip deflection in DLC 1.4 which agree with the results shown herein. Additionally, the maximum blade root bending moments in Ref. [1] are 90.5 MNm and 30.5 MNm for flapwise and edgewise, respectively. The results obtained in this work show maximum blade root bending moments of 86.6 MNm and 31.3 MNm for flapwise and edgewise, respectively. The maximum flapwise tip deflection in Ref. [1] is 22.8 m and in the analyses obtained herein maximum out-of-plane tip deflection is 23.5 m. The magnitude of the results shown in Fig. 7 generally agree well with the results reported in Ref. [1]. One potential source of the difference in loads that may benefit from further analysis is that waves were not included in the current analysis.



**Fig. 7 Maximum and minimum blade results from DLC 1.2, 1.4, and 1.5 simulations. Flapwise, edgewise, out-of-plane, and in-plane results are in accordance with OpenFAST coordinate systems for an upwind turbine.**

## VI. Baseline High-fidelity Structural Analysis of the UTD IEA 15 MW Blade

The UTD IEA 15 MW blade is analyzed in detail for deflection, strength, fatigue, buckling, and aero-elastic stability (modal results already presented in Section IV.C) in the following sub-sections.

### A. Deflection

It is important to evaluate the blade tip deflection in relation to the tower clearance. The DLC's analyzed herein correspond to normal operation, and thus a tower clearance of 30% must be met as stated in Section III.B. Equations 1- 3 shown below are used to calculate the tower clearance.

$$\text{Height of undeflected blade tip (m)} = Hub_h - R * \cos(C + U + A_{pb}) \quad (1)$$

$$\text{Blade tip clearance from tower axis (m)} = O * \cos(U) + R * \sin(C + U + A_{pb}) \quad (2)$$

$$\text{Blade tip clearance from tower (m)} = \text{Blade tip clearance from tower axis} + T_r \quad (3)$$

Where  $Hub_h$  is the hub height (150 m),  $R$  is the rotor radius (120 m),  $C$  is the blade pre-cone ( $-4^\circ$ ),  $U$  is the shaft up tilt ( $-6^\circ$ ),  $A_{pb}$  is the angle equivalent to the blade pre-bend ( $-1.91^\circ$ ),  $O$  is the overhang (-11.08 m), and  $T_r$  is the tower radius at the height of the undeflected blade tip. The  $T_r$  is determined using the tower dimensions specified in Ref. [1] at the height of the undeflected blade tip. Table 4 shows the result values for Eqs. 1- 3, and negative values indicate upwind distances.

**Table 4 Tower clearance results (negative values indicate upwind distances)**

|                                                 |   |        |
|-------------------------------------------------|---|--------|
| Height of undeflected blade tip                 | m | 32.58  |
| Blade tip clearance from tower axis             | m | -35.78 |
| Tower radius at height of undeflected blade tip | m | 4.99   |
| Blade tip clearance from tower                  | m | -30.79 |
| Admissible deflection (30% tower clearance)     | m | -21.56 |

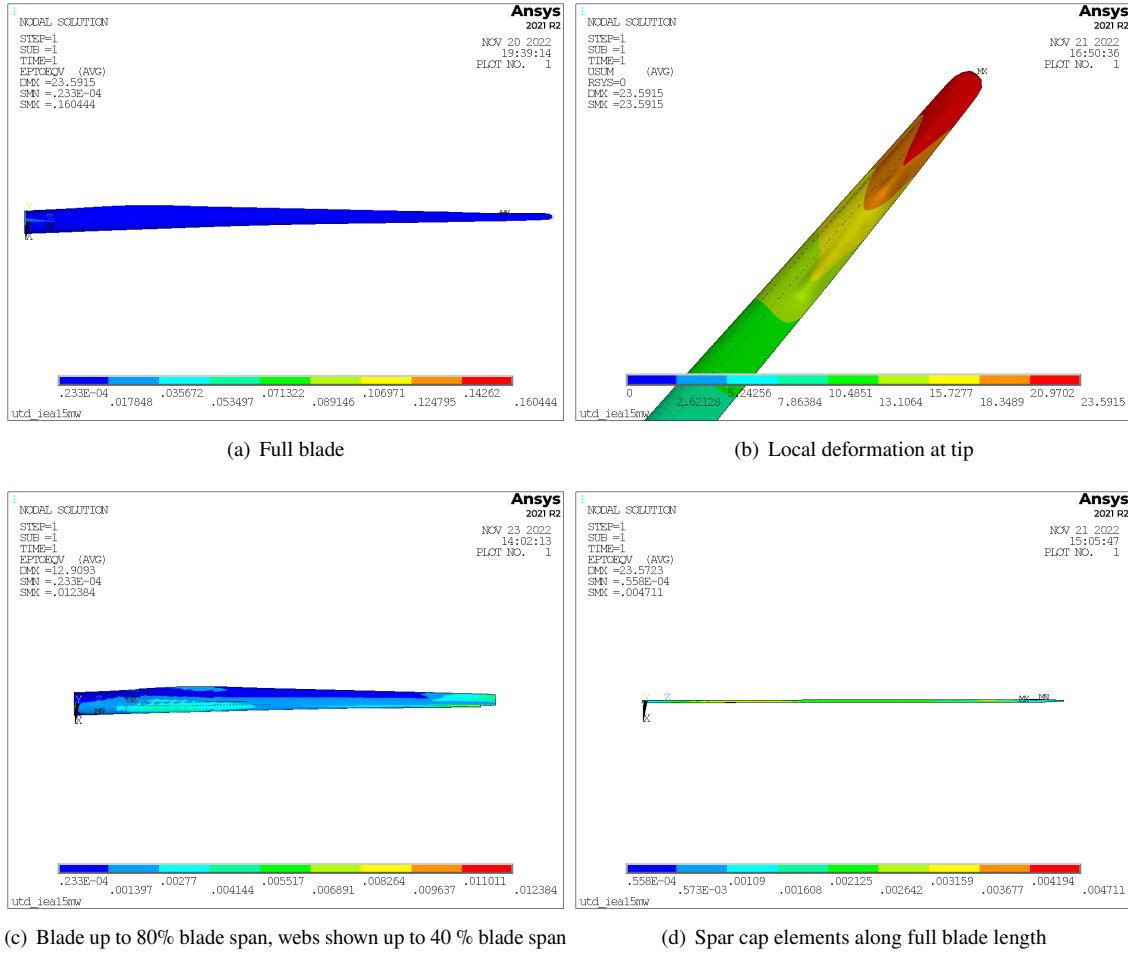
Figure 7 shows a maximum out-of-plane tip deflection of 23.49 m that occurs in DLC 1.4. This deflection exceeds the admissible deflection calculated above (21.56 m) by 1.93 m. However, the blade does not strike the tower since the blade tip clearance from the tower is 30.79 m.

### B. Strength and Fatigue

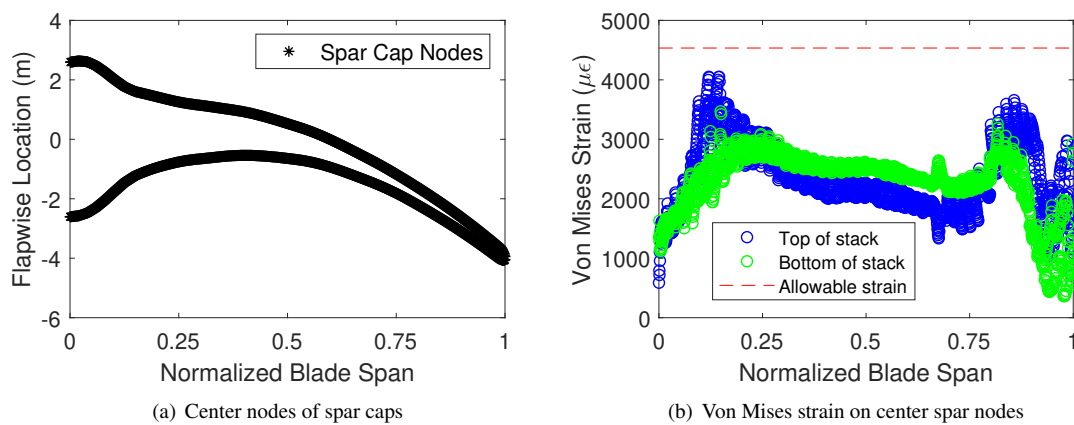
The strength performance is evaluated using ANSYS Mechanical APDL and strain results are shown in Fig. 8. The OpenFAST loads obtained in this work that result in the maximum combined root bending moment (90.4 MNm) correspond to DLC 1.4 as shown in Fig. 7, and these are converted to nodal loads and applied to the ANSYS blade model generated with AutoNuMAD. Using the safety factor for strength (2.977) presented in Section III.B, the tension allowable strains for tri-axial fiberglass (shell skins), bi-axial fiberglass (webs), and for carbon fiber (spar caps) are  $4635 \mu\epsilon$ ,  $1298 \mu\epsilon$ , and  $4535 \mu\epsilon$ , respectively.

The strains on the blade root, and on most of the trailing edge, leading edge, spar caps, and shear webs do not exceed the allowable strains as shown in Fig. 8(a) and Fig. 8(c). However, near max chord (around 20% of blade span) the leading edge has high strains and in the last 20% of the blade span the strains on the leading edge, trailing edge, and shear webs are high. Local deformations are observed in these areas as well as shown in Fig. 8(b) and these results are likely due to the trailing edge and leading edge reinforcements ending at 80% of the blade span and thin web skins. The IEA 15 MW design results in fractional number of layers that can work well for optimization and other methods that do not require high-fidelity modeling of the laminate; however, in this case fractional number of layers may aggravate the strains at the blade tip because in the conversion of the model from NuMAD to ANSYS the NuMAD tool rounds the number of layers to the nearest integer value (e.g., 0.4 layers would be rounded to zero layers resulting in missing structural material). The local tip deformations and high strains are easily solvable by having integer number of layers, adding more leading and trailing edge reinforcement layers, and increasing the laminate thickness.

Figure 8(d) shows the strain on the spar caps only and the allowable carbon strain is only exceeded by 3.88% near the tip. Figure 9 shows the Von Mises strain at the spar cap center axis (where all the layers are treated as a single stack and the strain results are shown at the top and bottom of the stack), and the strain results meet the allowable strain for carbon.



**Fig. 8 ANSYS Von Mises strain results.**

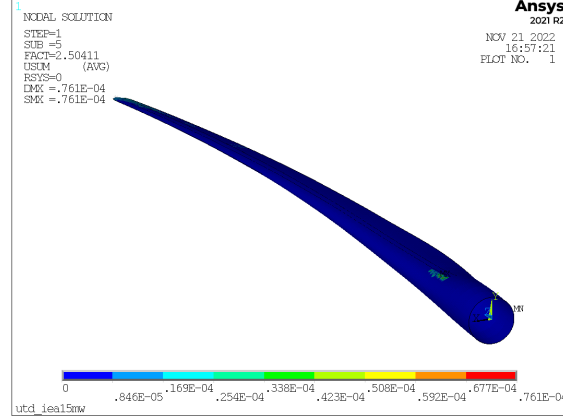


**Fig. 9 ANSYS Von Mises strain at center nodes of spar caps.**

The fatigue life of the blade is calculated with MLife using the results from the 72 simulations of DLC 1.2 described in Table 2. The calculated lifetime of the blade (flapwise and edgewise) exceeds 20 years, which is the minimum number of years required by the international design standards as stated in Section III.B.

### C. Buckling

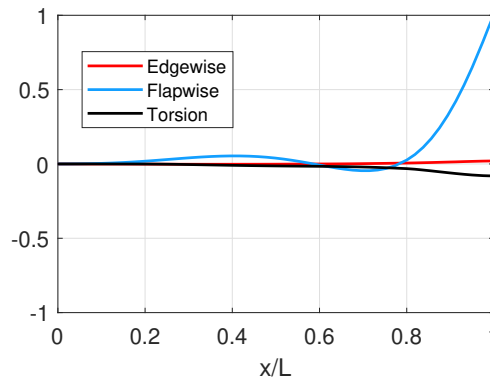
The same loads applied in the strength analyses are used to evaluate buckling. Figure 10 shows that buckling does not occur in the inboard section of the blade and close to the root until a buckling factor of 2.504 (exceeds 2.042 safety factor mentioned in Section III.B) which was determined by inspection of evaluating buckling 10 buckling modes for shift point eigenvalues of 0:0.5:2.5. However, the trailing edge buckles near the tip at factors much lower than the safety factor due to the thin structures present in this region that results in local deformations even in the static analyses as shown in Fig. 8(b).



**Fig. 10 ANSYS buckling result for lowest inboard buckling mode near root and forward of shear web.**

### D. Aero-Elastic Stability

The aero-elastic stability (specifically flutter) analysis uses the blade properties shown in Fig. 4 and the airfoil polars to determine the aero-elastic characteristics of the blade. Classical flutter is a dynamic instability that for wind turbine blades can be observed as a combination of torsional and flapwise mode shapes. The UTD IEA 15 MW blade shows a flutter speed of 6.87 rpm (0.91 flutter ratio compared to the 7.55 rated rpm of the turbine [1]) and the corresponding mode shapes is shown in Fig. 11. In addition, edgewise vibration instabilities were predicted below the rated rpm, having low damping similar to those observed in Ref. [24]. However, no instabilities were observed in the DLC results for the time-domain aero-servo-elastic simulations in OpenFAST. Further analysis may be required to investigate the aero-elastic stability of the IEA 15 MW blade in future studies that include high-fidelity blade structure re-design for flutter mitigation [24], and evaluation of flutter potential via time-domain aero-elastic simulations that consider aerodynamic damping present in operation.



**Fig. 11 Mode shape of classical coupled-mode flutter at 6.87 rpm**

## VII. Summary of Open-Source Files and Design Score Card

The main characteristics and structural performance results of the UTD IEA 15 MW blade are summarized in Table 5.

**Table 5 UTD IEA 15 MW blade model score card**

| Characteristic                                                      | Units         | Value                                                   |
|---------------------------------------------------------------------|---------------|---------------------------------------------------------|
| Power rating                                                        | MW            | 15                                                      |
| Wind class                                                          | -             | IB                                                      |
| Orientation                                                         | -             | Upwind                                                  |
| Number of blades                                                    | -             | 3                                                       |
| Rated wind speed                                                    | m/s           | 10.59                                                   |
| Hub radius                                                          | m             | 3                                                       |
| Blade length                                                        | m             | 117                                                     |
| Blade pre-cone                                                      | deg           | -4                                                      |
| Blade prebend                                                       | m             | 4                                                       |
| Shaft uptilt                                                        | deg           | -6                                                      |
| Blade mass                                                          | kg            | 68,077                                                  |
| 1st Flapwise parked frequency                                       | Hz            | 0.57                                                    |
| 1st Edgewise parked frequency                                       | Hz            | 0.65                                                    |
| Max. OpenFAST Out-of-plane deflection                               | m             | 23.5 (DLC 1.4)                                          |
| Operational admissible blade deflection<br>(assuming 30% clearance) | m             | 21.55                                                   |
| Max. Von Mises strain on spar caps                                  | $\mu\epsilon$ | 4711 near blade tip                                     |
| High strain regions                                                 | -             | Leading edge, trailing edge,<br>and webs near blade tip |
| Flapwise and edgewise fatigue life                                  | yrs           | >20                                                     |
| Buckling                                                            | ~             | 2.504 (near root)<br>< 1 (near tip)                     |
| Flutter ratio                                                       | ~             | 0.91                                                    |

All the UTD IEA 15 MW files are available in GitHub [8] for download. The files include the items marked in green in Fig. 1 and are described in Table 6.

## VIII. Conclusions

The work presented herein shows the conversion of the open-source IEA 15 MW offshore wind turbine blade v1.0 to NuMAD v2.0 format along with detailed high-fidelity structural analyses to baseline the blade structural performance. The UTD IEA 15 MW blade model can be a valuable resource that can be used in further optimization and research to help identify trends and solutions that drive towards technological innovations and reduction of cost. Modeling the blade in NuMAD offers many advantages some of which include access to an easily editable Excel file (where the blade geometry and internal structure are defined), and being connected to the high-fidelity software ANSYS Mechanical APDL that allows performing detail verification of the structural performance, such as strain and buckling.

The UTD IEA 15 MW blade model is available for download in GitHub [8]. A detailed description of the different blade modeling tools, and important differences in blade modeling between NuMAD v2.0 and WISDEM are provided in this work. Detailed structural evaluations on modes, deflection, strength, fatigue, buckling, and aero-elastic stability are performed considering all the international requirements for blade design and all results are documented in this work for reference. In future work, the NuMAD model developed herein will be used to perform a cost analysis using the cost model by Ref. [25] that interfaces with NuMAD.

**Table 6 UTD IEA 15 MW blade model files summary**

| Filename                                | Usage                                                                               | Description                                                                                                    |
|-----------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| NuMAD_utd_iea15mw.xlsx                  | Primary input file for NuMAD v2.0                                                   | Spreadsheet blade model data including detailed blade geometry, materials, layup information, and BOM summary. |
| utd_iea15mw.nmd                         | NuMAD model file                                                                    | Produced using NuMAD v2.0 with input from the NuMAD_utd_iea15mw.xlsx spreadsheet.                              |
| MatDBsi                                 | NuMAD materials database                                                            | Constraints material/laminate property information.                                                            |
| "airfoils" folder                       | NuMAD airfoil geometry coordinates                                                  | Contains set of files with coordinates for blade cross-section geometries.                                     |
| utd_iea15mw_ElastoDynBlade_ratedrpm.dat | Blade ElastoDyn input file used in OpenFAST simulations                             | OpenFAST blade file                                                                                            |
| shell7.src                              | Input file generated by NuMAD used to create the ANSYS model                        | Text file formatted for input to ANSYS to generate a finite element model                                      |
| utd_iea15mw.p3d                         | Blade external geometry file                                                        | Plot 3D file format                                                                                            |
| utd_iea15mw.iges                        | Geometry file exported from the ANSYS shell model that can be used in CAD modelling | Contains zero-thickness surfaces representing the outer moldline of the blade and the shear webs               |
| PreComp_SectionData.mat                 | Output MATLAB structure from PreComp                                                | Contains full set of cross-sectional blade properties generated by PreComp                                     |
| bmodes_0rpm.out                         | Output file from BModes at 0 rpm                                                    | Contains blade frequencies and mode shapes at 0 rpm (parked).                                                  |

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